



Dinesh Kumar Rajakumaran

Automotive Engineer | Mechatronics Engineer | Simulation Engineer | Systems Engineer |

MBSE | Testing & Validation

Contact

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GitHub

+43 677 61009477

LinkedIn

Wels, Austria

Portfolio

Engineering Skills

Teamwork

Problem-solving

Analytical thinking

Systems thinking

Cross-functional collaboration

Project coordination

Technical documentation

Technical Skills

MATLAB / Simulink

Python Programming

ROS / ROS Toolbox

Modelica

Real-Time Systems

Control Engineering (PID, MPC)

ISO 26262 / ISO 21434

SolidWorks

CATIA

Creo

ANSYS

Validation & Testing

Foxglove Studio

AUTOSAR

Languages

Tamil (Native)

English (C1 – Advanced)

German (B1 – Intermediate)

Professional Experience

Control Systems Engineering Intern

Oct 2024–Jun 2025

BRP-Rotax GmbH & Co KG, Gunskirchen, Austria

- Developed and validated nonlinear hydrofoil vehicle models in MATLAB/Simulink using state-space modelling, system linearization, and state estimation for control development.
- Designed and evaluated PID and MPC strategies under actuator constraints; analyzed closed-loop performance, robustness, and nonlinear behaviour across operating conditions.

Master's Project Student

Feb 2024–Jun 2024

Continental AG, Wels, Austria

- Optimised sensor-data processing using MATLAB ROS Toolbox and Foxglove Studio, developing custom Python-based message interfaces for ROS network integration.
- Configured IPv4/IPv6 ROS communication for real-time data exchange, enabling live visualization and validation of radar simulation signals.

Vehicle Testing Engineer

Mar 2022–Sep 2023

Royal Enfield Global Headquarters, Chennai, India

- Executed vehicle-level ABS and braking-system validation; analyzed development and regulatory test data against defined performance targets.
- Collaborated with Tier-1 suppliers including Bosch and Continental on system issues, test planning, and cross-functional validation throughout the vehicle development lifecycle.

Education

M.Sc. Automotive Mechatronics and Management

Sep 2023–Dec 2025

University of Applied Sciences Upper Austria, Wels, Austria

- Relevant Coursework:** Model-Based Engineering, Drivetrain Control, E-Mobility, Automotive Software Engineering, Automotive IT Systems, ADAS, Functional Safety (ISO 26262), Cybersecurity (ISO 21434), AUTOSAR, MATLAB/Simulink, Python, ROS-based Modelling & Simulation

B.E. Automobile Engineering

Jul 2017–Apr 2021

Rajalakshmi Engineering College, Chennai, India

- Relevant Coursework:** Vehicle Dynamics, Automotive Transmissions, Automotive Electronics & Embedded Systems, Control Engineering, Thermodynamics, Heat Transfer, Fluid Dynamics, Internal Combustion Engines, Theory of Machines, Manufacturing Technology